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Interview Preparation Short Notes

7 Chapters • 49 Topics • Quick-Reference Notes
Secure | Resilient | High-Performing | Cost-Optimized

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Chapter 1 — AWS Fundamentals, Security & IAM

1.1 Global Infrastructure & Well-Architected Framework

- Hierarchy: Region > Availability Zone (AZ) > Edge Location. A Region has 3+ isolated AZs; each AZ = one or more discrete data centers with redundant power/network.
- AZ names (us-east-1a) are randomized per account — the same AZ ID (use1-az1) maps to the same physical AZ across accounts.
- Edge Locations (400+) serve CloudFront/Route 53/Global Accelerator; Local Zones place compute near metros; Wavelength = 5G edge; Outposts = AWS hardware on-prem.
- Well-Architected 6 Pillars: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability.

TIP Design principle: stop guessing capacity, test at production scale, automate, allow for evolutionary architectures, drive from data, improve through game days.

- Shared Responsibility Model: AWS = security OF the cloud (hardware, global infra, managed service internals); Customer = security IN the cloud (data, IAM, OS patching on EC2, encryption config).

1.2 IAM Users / Groups / Policies

- IAM is global (not region-scoped). Root user should only be used for the few tasks that require it (close account, change support plan) — enable MFA on it and lock away keys.
- Users = long-term identities; Groups = collections of users (cannot nest groups, groups can't contain groups). Policies attach to users, groups, or roles.
- Policy = JSON with Effect (Allow/Deny), Action, Resource, optional Condition. Explicit Deny always wins over any Allow.

```
{ "Effect": "Allow", "Action": "s3:GetObject",  
  "Resource": "arn:aws:s3:::my-bucket/*",  
  "Condition": { "IpAddress": { "aws:SourceIp": "10.0.0.0/16" } } }
```

- Managed policies (AWS-managed or customer-managed, reusable, versioned) vs inline policies (1:1 embedded, deleted with the identity).
- Evaluation logic: default Deny → evaluate all applicable policies → explicit Allow grants → explicit Deny overrides. SCPs, permission boundaries, session policies further restrict.

TIP Least privilege: start minimal, use IAM Access Analyzer + last-accessed data to right-size.

1.3 IAM Roles & Advanced IAM

- Roles = temporary credentials via STS (AssumeRole), no long-term keys. Use for EC2 (instance profile), Lambda, cross-account access, and federation.
- Trust policy (who can assume) vs permission policy (what they can do). Cross-account = trust the other account's ID + external ID to prevent the confused-deputy problem.
- STS APIs: AssumeRole, AssumeRoleWithSAML (enterprise SSO), AssumeRoleWithWebIdentity (Cognito/OIDC — prefer Cognito for mobile).
- Permission Boundary = max permissions an identity can have (guardrail); effective perms = intersection of boundary AND identity policy.
- IAM Identity Center (successor to AWS SSO) centralizes workforce access across accounts with permission sets.

WATCH EC2 credentials come from instance metadata (IMDS). Enforce IMDSv2 (session token, hop limit) to block SSRF credential theft.

1.4 AWS Organizations & Multi-Account Strategy

- Organization = management (payer) account + member accounts, grouped into OUs (up to nested hierarchy). Enables consolidated billing + volume discounts + shared Reserved Instances/Savings Plans.
- Service Control Policies (SCPs) set permission guardrails on OUs/accounts — they do NOT grant permissions, only limit the maximum. Do not apply to the management account.
- SCP + IAM = effective permissions are the intersection. An action must be allowed by both SCP and IAM policy.
- Multi-account benefits: blast-radius isolation, per-account billing/quotas, separate environments (prod/dev), security boundary.
- Control Tower = automated landing zone (guardrails, account factory, centralized logging to a Log Archive account, Audit account).

TIP Tag policies + AWS Config + centralized CloudTrail (org trail) give org-wide governance and audit.

1.5 Data Encryption & Key Management (KMS)

- Encryption at rest (data on disk) vs in transit (TLS). KMS manages symmetric (default, AES-256) and asymmetric keys; keys never leave KMS unencrypted.
- KMS key types: AWS-managed (aws/service), Customer-Managed Keys (CMK — you control policy/rotation), AWS-owned. Automatic annual rotation for CMKs (optional, 1-year).
- Envelope encryption: KMS encrypts a data key; the data key encrypts the data. GenerateDataKey returns plaintext + encrypted key.

- KMS key policy is the primary access control (resource-based); combine with IAM and grants. Cross-account use requires key policy + IAM.
- CloudHSM = single-tenant, FIPS 140-2 Level 3 dedicated HSM (you manage keys). Use when regulatory control of the HSM is required.

WATCH KMS has request throttling (per-region). High-throughput crypto → use data-key caching or S3 bucket keys to reduce KMS calls/cost.

1.6 AWS Security Services

- GuardDuty = intelligent threat detection (analyzes CloudTrail, VPC Flow Logs, DNS logs) — no agents.
- Macie = ML-based sensitive data (PII) discovery in S3.
- Inspector = automated vulnerability scanning for EC2, ECR images, Lambda.
- Security Hub = aggregates findings across GuardDuty/Inspector/Macie + compliance standards (CIS, PCI-DSS).
- WAF = layer-7 protection (SQLi, XSS, rate limiting) on ALB/CloudFront/API Gateway. Shield Standard = free DDoS; Shield Advanced = paid, 24/7 DRT + cost protection.
- Detective = root-cause investigation; Config = resource configuration compliance + change history.

TIP Interview mnemonic: GuardDuty detects, Inspector scans, Macie classifies, WAF filters, Shield absorbs DDoS.

1.7 Secrets Management & Secure Application Design

- Secrets Manager: stores DB creds/API keys, supports automatic rotation (built-in for RDS/Aurora/Redshift/DocumentDB via Lambda). Costs per secret + API calls.
- SSM Parameter Store: free standard tier for config + secrets (SecureString via KMS); no native rotation but cheaper. Advanced tier for larger/more params.

TIP Choose Secrets Manager when you need automatic rotation & cross-account sharing; Parameter Store for simple config at no cost.

- Never hardcode credentials — use IAM roles for AWS service auth, Secrets Manager/Parameter Store for third-party creds.
- Defense in depth: encrypt at rest + in transit, least-privilege IAM, private subnets, security groups, and rotate credentials.

Chapter 2 — Compute Services

2.1 EC2 Instance Types & Purchasing Options

- Families: T (burstable), M (general), C (compute), R/X (memory), I/D (storage), P/G/Inf (accelerated/GPU/ML). Naming: m5.large = family/gen/size.
- Purchasing: On-Demand (pay/second, no commit), Reserved Instances (1/3-yr, up to 72% off — Standard vs Convertible), Savings Plans (flexible commit \$/hr — Compute vs EC2 Instance), Spot (up to 90% off, interruptible with 2-min warning), Dedicated Hosts (compliance/BYOL), Dedicated Instances.
- Spot use cases: fault-tolerant, stateless, batch, big data. Spot Fleet / mixed instances policy diversify across pools. Interruption behaviors: terminate/stop/hibernate.

TIP Savings Plans > RIs for most cases due to flexibility. Compute Savings Plans apply across regions, families, even Fargate/Lambda.

- Capacity Reservations guarantee capacity in an AZ (billed regardless of use); combine with Savings Plans for discount.
- T-family CPU credits: baseline + burst. T2/T3 Unlimited mode allows sustained burst (extra cost) vs Standard mode throttling.

2.2 EC2 Networking, Storage & AMIs

- ENI = virtual NIC (primary/secondary private IPs, EIP, SG). Enhanced networking: ENA (up to 100 Gbps), EFA for HPC/MPI. Placement groups: Cluster (low latency), Spread (hardware isolation, max 7/AZ), Partition (large distributed like Hadoop).
- Storage: EBS (network block, persists), Instance Store (physical, ephemeral, high IOPS, lost on stop/terminate).
- AMI = template (OS + config + apps). Backed by EBS snapshot. Can copy across regions/accounts; share or make public.
- User data runs at first boot (bootstrap). Instance metadata service (IMDS) exposes instance info + role creds at 169.254.169.254.

TIP Golden AMI pattern: bake dependencies into a custom AMI (via EC2 Image Builder) for faster, consistent launches.

2.3 EC2 Auto Scaling & High Availability

- Auto Scaling Group (ASG): min / desired / max capacity across multiple AZs (spread for HA). Uses Launch Template (preferred) or Launch Config (legacy).
- Scaling policies: Target Tracking (keep a metric e.g. 50% CPU), Step Scaling, Simple Scaling, Scheduled (predictable load), Predictive (ML forecast).

- Health checks: EC2 (status checks) or ELB (application-level). Cooldown prevents thrashing. Lifecycle hooks pause instances for setup/teardown.
- For HA: ASG across ≥ 2 AZs behind an ELB; ELB routes only to healthy targets and replaces failed instances automatically.

WATCH Scale-in protection prevents ASG from terminating specific instances during scale-in.

2.4 AWS Lambda & Serverless Computing

- Event-driven, no server management, pay per request + GB-seconds. Max timeout 15 min; memory 128 MB–10 GB (CPU scales with memory).
- Concurrency: default 1000/region (soft limit); Reserved concurrency caps a function; Provisioned concurrency pre-warms to avoid cold starts.
- Triggers: API Gateway, S3, DynamoDB Streams, SQS, SNS, EventBridge, Kinesis. Sync (request/response) vs async (retries + DLQ) vs poll-based (stream/queue).
- Cold start factors: runtime, package size, VPC ENI attach. Reduce with smaller packages, provisioned concurrency, and keeping functions warm.
- /tmp ephemeral storage 512 MB–10 GB; use Lambda layers for shared deps; attach to VPC only when it needs private resources.

TIP Lambda in a VPC now uses Hyperplane ENIs — shared, so cold-start ENI penalty is largely gone.

2.5 API Gateway

- Managed API front door: REST, HTTP (cheaper/faster, fewer features), and WebSocket APIs. Handles throttling, auth, caching, versioning (stages).
- Auth options: IAM (SigV4), Cognito user pools, Lambda authorizers (custom token/request), API keys + usage plans (rate/quota per client).
- Integrations: Lambda proxy, HTTP, AWS service, mock. Caching at the stage level reduces backend calls (TTL configurable).
- Throttling: account-level + per-method; returns 429 when exceeded. Use usage plans to protect the backend.

TIP Choose HTTP API for simple Lambda/HTTP proxies (lower cost/latency); REST API for advanced features (caching, request validation, WAF, private APIs).

2.6 Container Services (ECS & EKS)

- ECS = AWS-native orchestration. Launch types: EC2 (you manage instances) or Fargate (serverless, no instance management). Task definition = container blueprint; Service maintains desired count.

- EKS = managed Kubernetes control plane; run on EC2 or Fargate. Choose EKS for Kubernetes portability/ecosystem; ECS for simpler AWS-integrated workloads.
- ECR = private container registry (integrates with IAM, image scanning). Fargate bills per vCPU/GB-second.
- Task IAM roles give containers scoped AWS permissions. ALB integrates with ECS/EKS for dynamic port mapping + target groups.

TIP Fargate = no EC2 management, good for spiky/unpredictable; EC2 launch type = cheaper at steady high utilization + GPU/custom needs.

2.7 Elastic Beanstalk & Additional Compute

- Elastic Beanstalk = PaaS: upload code, it provisions EC2/ASG/ELB/RDS. You keep full control of resources. Free (pay for underlying resources).
- Deployment policies: All-at-once (downtime), Rolling, Rolling with additional batch, Immutable (new instances), Blue/Green (swap URLs via Route 53).
- AWS Batch = managed batch computing (job queues on EC2/Fargate/Spot). Lightsail = simplified VPS with predictable pricing.
- App Runner = fully managed containers from source/image (no infra). Outposts/Wavelength/Local Zones = edge/hybrid compute.

TIP Beanstalk for developers who want quick deploys without managing infra but still need visibility; App Runner for even simpler web apps/APIs.

Chapter 3 — Storage Services

3.1 Amazon S3 Core Concepts & Storage Classes

- Object storage (not a filesystem). Bucket names globally unique; objects up to 5 TB; keys are flat (prefixes simulate folders). 11 9s durability across ≥ 3 AZs.
- Storage classes: Standard (frequent), Intelligent-Tiering (auto-move, no retrieval fee), Standard-IA (infrequent, retrieval fee), One Zone-IA (single AZ, cheaper, less durable), Glacier Instant Retrieval, Glacier Flexible Retrieval (mins–hrs), Glacier Deep Archive (12 hr, cheapest).
- Lifecycle rules transition objects between classes + expire them. Versioning keeps object history (protects against deletes/overwrites).

TIP Intelligent-Tiering = default choice when access patterns are unknown/changing — no retrieval charges, small monitoring fee.

- Strong read-after-write consistency for all operations (since Dec 2020). Multipart upload for objects >100 MB (required >5 GB).

3.2 S3 Security & Access Management

- Access controls: IAM policies (identity), bucket policies (resource, JSON), ACLs (legacy, avoid), and Block Public Access (account/bucket, ON by default).
- Encryption: SSE-S3 (AES-256, AWS keys), SSE-KMS (audit + rotation, KMS costs), SSE-C (customer-provided keys), or client-side. Default encryption can be enforced.
- Pre-signed URLs grant temporary object access (inherit signer's permissions). VPC Gateway Endpoint keeps S3 traffic private (no NAT/IGW).
- S3 Access Points simplify shared-bucket access at scale (per-app policies + network origin controls).

WATCH Public S3 buckets are a top breach cause — keep Block Public Access ON; use bucket policies + CloudFront OAC for controlled public content.

3.3 S3 Advanced Features & Performance

- Performance scales to 3,500 PUT/COPY/POST/DELETE and 5,500 GET/HEAD per prefix per second — parallelize across prefixes.
- Transfer Acceleration uses CloudFront edge for faster long-distance uploads. Multipart upload parallelizes large files.
- S3 Replication: CRR (cross-region — DR/latency/compliance) and SRR (same-region — log aggregation/prod-to-test). Needs versioning on both buckets.
- Event notifications → Lambda/SQS/SNS/EventBridge. S3 Object Lambda transforms data on GET. Requester Pays shifts cost to requester.

- Object Lock (WORM) for compliance retention; MFA Delete adds protection to versioned deletes. S3 Select queries CSV/JSON/Parquet in place.

3.4 Amazon EBS

- Network-attached block storage for EC2; AZ-scoped (attach only within same AZ). Persists independently of instance (unless delete-on-termination).
- Volume types: gp3 (baseline 3000 IOPS/125 MB/s, independently scalable — default SSD), gp2 (IOPS tied to size), io1/io2 (high IOPS, io2 Block Express to 256K IOPS), st1 (throughput HDD — big data), sc1 (cold HDD — archive).
- Snapshots = incremental, stored in S3, regional. Copy across regions/accounts; create AMIs from them. Fast Snapshot Restore removes first-touch latency.
- Encryption via KMS (encrypts volume, snapshots, and data in transit between instance and volume). Multi-Attach (io1/io2) allows one volume on multiple instances in same AZ.

TIP gp3 is cheaper than gp2 and lets you provision IOPS/throughput separately — default recommendation for most workloads.

3.5 Amazon EFS & FSx

- EFS = managed NFS (Linux), elastic, multi-AZ, mounts on many EC2 concurrently. Performance modes: General Purpose vs Max I/O; Throughput: Bursting vs Provisioned/Elastic. Storage classes: Standard + IA (lifecycle).
- FSx for Windows File Server = SMB/NTFS/AD-integrated shares. FSx for Lustre = HPC/ML high-throughput, S3-linked. FSx for NetApp ONTAP + OpenZFS = multi-protocol/enterprise features.

TIP EFS = Linux shared file; FSx Windows = Windows/SMB; FSx Lustre = HPC + S3; EBS = single-instance block; S3 = object.

- EFS scales to petabytes automatically, pay per GB used; access points enforce per-app root directory + POSIX identity.

3.6 Hybrid Storage Solutions (Storage Gateway)

- AWS Storage Gateway bridges on-prem to AWS. Types: File Gateway (NFS/SMB → S3), Volume Gateway (iSCSI: Cached = primary in S3/local cache, Stored = primary on-prem/async backup to S3), Tape Gateway (VTL → Glacier).
- DataSync = fast one-time or scheduled bulk transfer (on-prem ↔ AWS or AWS ↔ AWS), handles NFS/SMB/S3/EFS/FSx with encryption + validation.

TIP Storage Gateway = ongoing hybrid access; DataSync = migration/replication of large datasets.

- Use File Gateway for on-prem apps needing S3-backed file access with local caching; Tape Gateway to replace physical tape backups.

3.7 AWS Snow Family

- Offline/edge data transfer + compute. Snowcone (8/14 TB, smallest, portable, DataSync agent), Snowball Edge (Storage Optimized ~80 TB or Compute Optimized w/ GPU), Snowmobile (exabyte-scale, a truck — being retired for most).
- Use when network transfer would take too long (rule of thumb: >1 week over the wire → consider Snow). Data encrypted with KMS; chain-of-custody + tamper-evident.

TIP Calculate: transferring 100 TB over 1 Gbps $\approx$ 12+ days — Snowball ships in days. Bandwidth math is a common exam trigger.

- Snowball Edge also runs EC2/Lambda at disconnected edge sites (ships, factories, military).

Chapter 4 — Database Services

4.1 Amazon RDS

- Managed relational DB: MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, + Aurora. AWS handles patching, backups, HA. You don't get OS/SSH access.
- Multi-AZ = synchronous standby in another AZ for HA/failover (not for reads) — automatic DNS failover. Read Replicas = asynchronous, scale reads, can be cross-region, promotable.
- Backups: automated (point-in-time recovery, 1–35 day retention) + manual snapshots (kept until deleted). Restores create a NEW instance.
- Storage: gp3/io1 with autoscaling. Encryption at rest via KMS (must enable at creation; snapshot+copy to encrypt existing). TLS in transit.

TIP Multi-AZ = availability; Read Replica = scalability. Both together for read-heavy HA. RDS Proxy pools connections (great for Lambda).

- RDS Custom (Oracle/SQL Server) allows OS/DB customization. Blue/Green Deployments for low-risk upgrades.

4.2 Amazon Aurora

- MySQL/PostgreSQL-compatible, 5x MySQL / 3x PostgreSQL throughput. Storage auto-grows to 128 TB, 6 copies across 3 AZs, self-healing.
- Up to 15 read replicas with ~sub-10ms replica lag (shared storage layer). Writer + reader endpoints; custom endpoints for workload routing.
- Aurora Serverless v2 = fine-grained auto-scaling ACUs for variable workloads. Global Database = cross-region replication (<1s), <1 min RTO for DR.
- Backtrack (rewind cluster without restore), fast clones (copy-on-write), parallel query. Multi-master (write to multiple nodes) for select cases.

TIP Choose Aurora over RDS for higher performance, faster failover (~30s), and cloud-native DR; RDS for other engines (Oracle/SQL Server) or lower cost.

4.3 Amazon DynamoDB Fundamentals

- Serverless NoSQL key-value/document; single-digit ms latency at any scale. Fully managed, multi-AZ by default, no servers.
- Primary key: Partition key (hash) or Partition + Sort key (composite). Items ≤400 KB. Data auto-partitioned by partition key hash.
- Capacity modes: On-Demand (pay per request, auto-scale, spiky/unknown) vs Provisioned (RCU/WCU, cheaper for steady, with auto-scaling).

- 1 RCU = 1 strongly-consistent read/s of 4 KB (2 eventually-consistent). 1 WCU = 1 write/s of 1 KB. Design keys to avoid hot partitions.

WATCH Poor partition-key choice → hot partitions + throttling. Use high-cardinality keys and distribute access evenly.

4.4 DynamoDB Advanced Features

- Secondary indexes: GSI (different partition/sort key, own capacity, eventually consistent) vs LSI (same partition key, alt sort key, must create at table creation, strong consistency).
- DynamoDB Streams = ordered change log (24 hr) → trigger Lambda for event-driven pipelines/replication. TTL auto-deletes expired items.
- DAX = in-memory cache (microsecond reads) for read-heavy/repeated queries. Global Tables = multi-region, multi-active replication.
- Transactions (ACID) across items/tables. PITR (35-day point-in-time recovery) + on-demand backups. Optimistic locking with version numbers.

TIP GSI when you need to query on a non-key attribute; DAX for microsecond read caching; Global Tables for multi-region active-active.

4.5 Amazon ElastiCache

- Managed in-memory cache: Redis (or Valkey) and Memcached. Offloads DB reads, reduces latency, stores sessions.
- Redis: persistence, replication (read replicas), Multi-AZ + auto failover, backups, pub/sub, sorted sets, cluster mode sharding. Choose for HA/rich data structures.
- Memcached: simple, multi-threaded, horizontally scalable, no persistence/replication. Choose for simple caching where data loss is acceptable.
- Caching strategies: Lazy loading (cache on read miss) vs Write-through (write to cache+DB). TTL prevents stale data.

TIP Redis = HA + advanced structures + persistence; Memcached = simple, multi-threaded scaling. Session store → Redis.

4.6 Amazon Redshift & Analytics

- Petabyte-scale columnar data warehouse (OLAP), based on PostgreSQL. Massively parallel processing (MPP) across nodes; columnar storage + compression for analytics.
- Leader node (query planning) + compute nodes (slices). RA3 nodes = managed storage (scale compute/storage independently) with Redshift Managed Storage.

- Redshift Spectrum queries S3 data directly (no load). Concurrency Scaling adds transient clusters for spikes. Serverless option auto-provisions.
- Loads via COPY from S3; integrates with Kinesis, Glue, QuickSight. Distribution styles (KEY/EVEN/ALL) + sort keys tune performance.

TIP Redshift = complex analytical queries on structured data at scale (OLAP); Athena = serverless ad-hoc SQL on S3; not for OLTP.

4.7 Purpose-Built Databases

- DocumentDB = managed MongoDB-compatible document DB. Keyspaces = managed Cassandra. Neptune = graph DB (social, fraud, recommendations).
- Timestream = time-series (IoT, metrics). QLDB = immutable, cryptographically verifiable ledger. MemoryDB = Redis-compatible durable in-memory primary DB.
- ElastiCache vs MemoryDB: ElastiCache = cache alongside a DB; MemoryDB = durable primary DB with in-memory speed.

TIP Match workload to engine: relational→RDS/Aurora, key-value→DynamoDB, cache→ElastiCache, graph→Neptune, time-series→Timestream, ledger→QLDB, warehouse→Redshift, search→OpenSearch.

Chapter 5 — Networking & Content Delivery

5.1 VPC Fundamentals

- VPC = isolated virtual network in a region (CIDR /16–/28, RFC1918 private ranges). Subnets are AZ-scoped; public subnet = route to Internet Gateway.
- Internet Gateway (IGW) = internet access for public subnets. NAT Gateway (managed, AZ-scoped, HA within AZ) lets private subnets reach internet outbound only.
- Route tables direct traffic (local route always present). Each subnet associates with one route table. Reserve 5 IPs per subnet (first 4 + last).
- DNS: enableDnsSupport + enableDnsHostnames. Secondary CIDRs can extend a VPC. DHCP option sets customize DNS/domain.

TIP Public subnet has a route to IGW; private subnet routes internet-bound traffic to a NAT Gateway (in a public subnet).

5.2 VPC Security

- Security Groups = stateful, instance/ENI level, allow rules only (implicit deny), return traffic auto-allowed. Reference other SGs.
- Network ACLs = stateless, subnet level, allow AND deny rules, evaluated in order (rule #), return traffic needs explicit rule. Default NACL allows all.

TIP SG = stateful + allow-only (first line, per-instance); NACL = stateless + allow/deny (subnet-wide, e.g. block a malicious IP).

- VPC Flow Logs capture IP traffic metadata (accepted/rejected) to CloudWatch/S3 for troubleshooting + security analysis.
- Layered defense: NACL (subnet) → SG (instance) → host firewall. Use SG references for tier-to-tier (web→app→db) rules.

5.3 VPC Connectivity & Endpoints

- VPC Peering = 1:1 private connection between VPCs (non-transitive, no overlapping CIDRs). Cross-region/account supported.
- VPC Endpoints keep traffic on the AWS network (no IGW/NAT). Gateway Endpoints (S3, DynamoDB — free, route-table entry) vs Interface Endpoints (PrivateLink, ENI + hourly cost, most services).
- Transit Gateway = hub-and-spoke connecting many VPCs + on-prem (transitive routing) — replaces mesh peering at scale.
- PrivateLink exposes a service privately across VPCs/accounts via interface endpoints (no peering, no exposed CIDR).

TIP S3/DynamoDB → Gateway Endpoint (free). Other services / on-prem private access → Interface Endpoint (PrivateLink).

5.4 Hybrid Connectivity

- Site-to-Site VPN = encrypted IPsec tunnel over internet (quick to set up, variable performance). Two tunnels for redundancy.
- Direct Connect (DX) = dedicated private physical link to AWS (consistent low latency, higher bandwidth, lower egress cost). Takes weeks to provision.
- DX + VPN = encrypted DX (best of both). Multiple DX connections/locations for resilience. Public VIF (public services) vs Private VIF (VPC) vs Transit VIF (Transit Gateway).

TIP Need it fast/cheap → VPN. Need consistent performance/bandwidth/compliance → Direct Connect (with VPN backup for HA).

- Transit Gateway + DX Gateway centralizes hybrid connectivity across many VPCs/regions.

5.5 Elastic Load Balancing (ELB)

- ALB = layer 7 (HTTP/HTTPS), host/path-based routing, WebSockets, containers, Lambda targets, WAF integration. Best for web apps/microservices.
- NLB = layer 4 (TCP/UDP/TLS), ultra-low latency, millions of req/s, static IP + Elastic IP per AZ. Best for extreme performance/static IP.
- GWLB = layer 3 for deploying third-party virtual appliances (firewalls/IDS). CLB = legacy (avoid for new designs).
- Features: health checks, cross-zone load balancing, connection draining/deregistration delay, sticky sessions, SSL termination + SNI.

TIP ALB=HTTP smart routing; NLB=TCP/UDP + static IP + extreme scale; GWLB=inline security appliances.

5.6 Amazon Route 53

- Managed DNS + domain registration + health checks. 100% SLA. Record types: A, AAAA, CNAME, Alias (AWS-specific, free, points to ELB/CloudFront/S3, works at zone apex).
- Routing policies: Simple, Weighted (A/B, canary), Latency (lowest latency region), Failover (active-passive w/ health checks), Geolocation (by user location), Geoproximity (traffic bias), Multivalue Answer (up to 8 healthy records).
- Health checks monitor endpoints and enable DNS failover. Private hosted zones resolve within VPCs.

TIP Alias > CNAME for apex domains + free + AWS-target aware. Latency routing = performance; Geolocation = compliance/localization.

5.7 Amazon CloudFront & Global Accelerator

- CloudFront = global CDN caching content at edge locations (lower latency). Origins: S3, ALB, EC2, any HTTP. Origin Access Control (OAC) restricts S3 to CloudFront only.
- Features: HTTPS/TLS, field-level encryption, signed URLs/cookies (restrict access), Lambda@Edge & CloudFront Functions (edge compute), geo-restriction, WAF + Shield integration, custom cache behaviors + TTLs.
- Global Accelerator = uses AWS backbone + anycast static IPs to route TCP/UDP to nearest healthy endpoint (non-cacheable/dynamic traffic, gaming, VoIP, fast failover).

TIP CloudFront = cache static/web content (HTTP); Global Accelerator = improve dynamic/non-HTTP traffic + provides 2 static anycast IPs + fast regional failover.

Chapter 6 — Designing Resilient & High-Performing Architectures

6.1 High Availability & Fault Tolerance

- HA = minimize downtime (redundancy, failover). Fault tolerance = keep running through failures (no impact). Multi-AZ = HA within region; Multi-Region = DR + global HA.
- Eliminate single points of failure: multiple AZs, ASG, ELB, Multi-AZ RDS, stateless app tier (externalize sessions to DynamoDB/ElastiCache).
- Loose coupling with SQS/SNS decouples components so a failure isn't cascading. Health checks + auto-recovery replace failed resources.

TIP Design for failure: assume everything fails; use redundancy, health checks, automation, and graceful degradation.

6.2 Disaster Recovery Strategies

- Four strategies by RTO/RPO/cost: Backup & Restore (cheapest, hours RTO), Pilot Light (core minimal running, scale on failover), Warm Standby (scaled-down full copy, faster), Multi-Site Active/Active (near-zero RTO/RPO, most expensive).
- RTO = time to recover; RPO = acceptable data loss window. Lower RTO/RPO = higher cost.
- Tools: AWS Backup (centralized), cross-region snapshots/AMIs, Aurora Global DB, DynamoDB Global Tables, Route 53 failover, S3 CRR, Elastic Disaster Recovery (DRS).

TIP Match strategy to RTO/RPO: Backup&Restore (hrs), Pilot Light (10s min), Warm Standby (mins), Active/Active (near-zero).

6.3 Decoupled & Event-Driven Architectures

- SQS = fully managed message queue (pull, decouple, buffer). Standard (at-least-once, best-effort order, high throughput) vs FIFO (exactly-once, ordered, 300–3000 msg/s). Visibility timeout + DLQ for poison messages.
- SNS = pub/sub fan-out (push to multiple subscribers: SQS, Lambda, HTTP, email, SMS). SNS+SQS fan-out pattern = durable multi-consumer.
- EventBridge = event bus with schema registry, rules, SaaS integrations, and scheduling (replaces CloudWatch Events). Kinesis = real-time streaming (Data Streams/Firehose/Analytics).

TIP Queue (SQS) = decouple + buffer one consumer; Topic (SNS) = fan-out to many; EventBridge = routing/filtering across many sources; Kinesis = high-volume ordered streaming + replay.

- Idempotency + DLQs make async processing resilient; long polling reduces empty SQS receives + cost.

6.4 Auto Scaling Strategies

- Scale EC2 (ASG), DynamoDB, Aurora replicas, ECS services, and more via Application Auto Scaling. Match policy to pattern.
- Target tracking = simplest (maintain a metric). Scheduled = known peaks (business hours). Predictive = ML forecasts for proactive scaling. Step = graduated response to alarm magnitude.
- Scale out early, scale in slowly (cooldowns) to avoid flapping. Warm pools pre-initialize instances to cut launch latency.

TIP Combine predictive (baseline) + dynamic target-tracking (real-time correction) for spiky, partly-predictable load.

6.5 Performance Optimisation — Compute & Storage

- Right-size instances (Compute Optimizer recommendations). Choose the right family (C for CPU, R for memory, etc.). Use Graviton (ARM) for better price/performance.
- Caching layers: CloudFront (edge), ElastiCache (DB), DAX (DynamoDB), API Gateway cache — reduce load + latency.
- Storage perf: gp3/io2 for IOPS, instance store for ephemeral high IOPS, EFS Max I/O / FSx Lustre for parallel throughput, S3 prefix parallelism + Transfer Acceleration.

TIP Add caching before scaling out; measure with CloudWatch; use Compute Optimizer + Trusted Advisor to right-size.

6.6 Performance Optimisation — Database & Network

- DB: read replicas (scale reads), ElastiCache/DAX (cache), RDS Proxy (connection pooling for serverless), proper indexing + partition keys, Aurora for throughput.
- Choose the right DB per workload (OLTP vs OLAP vs key-value) — don't force relational for everything.
- Network: CloudFront + Global Accelerator (edge/backbone), placement groups (cluster low-latency), enhanced networking (ENA/EFA), Direct Connect for consistent hybrid latency, VPC endpoints to avoid internet hops.

TIP Reduce round trips (cache), reduce distance (edge/backbone), reduce contention (read replicas + connection pooling).

6.7 Monitoring & Observability

- CloudWatch = metrics, logs, alarms, dashboards, custom metrics, Logs Insights queries. Alarms trigger scaling/SNS/actions. Detailed monitoring = 1-min granularity.

- CloudTrail = API audit log (who did what, when) — governance/compliance/forensics. Multi-region org trail to S3 + CloudWatch Logs.
- AWS Config = resource config compliance + change history + rules. X-Ray = distributed tracing for microservices bottlenecks.

TIP CloudWatch = performance/health (what's happening); CloudTrail = audit (who did it); Config = compliance (is it configured correctly).

- EventBridge + CloudWatch Alarms automate remediation; centralize logs for cross-account observability.

Chapter 7 — Cost Optimisation & Exam Preparation

7.1 AWS Pricing Models

- Core levers: pay-as-you-go, commit for discount (RIs/Savings Plans), pay less by reserving capacity, and pay less as AWS grows (price drops).
- Compute: On-Demand > Savings Plans/RIs (up to 72% off) > Spot (up to 90% off). Storage tiers (S3 classes, EBS gp3) reduce cost by access pattern.
- Data transfer: inbound usually free; outbound to internet costs; cross-AZ/region costs; use CloudFront/VPC endpoints to reduce egress.

TIP Biggest savings: commitment discounts (Savings Plans) + Spot for fault-tolerant + right-sizing + storage tiering + reducing data transfer.

7.2 Cost Optimisation Strategies

- Right-size (Compute Optimizer), delete idle/unattached resources (EIPs, EBS, old snapshots), schedule non-prod shutdowns, and use auto scaling to match demand.
- Use Spot for batch/CI/big data, Savings Plans/RIs for steady baseline, and serverless (Lambda/Fargate) to pay only for use.
- Storage: S3 Intelligent-Tiering + lifecycle rules, gp3 over gp2, delete orphaned snapshots. Serverless DBs (Aurora Serverless, DynamoDB On-Demand) for variable load.

TIP Tag everything for cost allocation; review Trusted Advisor + Cost Explorer regularly; adopt FinOps practices.

7.3 Cost Management Tools

- Cost Explorer = visualize + forecast spend, find trends, RI/SP recommendations. AWS Budgets = alerts on cost/usage/RI thresholds (+ actions).
- Cost & Usage Report (CUR) = most granular billing data (to S3, query with Athena/QuickSight). Trusted Advisor = cost/security/perf/fault-tolerance/limit checks.
- Compute Optimizer = right-sizing recommendations. Billing alarms via CloudWatch. Cost Anomaly Detection flags unexpected spikes.

TIP Budgets = proactive alerts; Cost Explorer = analysis; CUR = deep dive; Trusted Advisor = quick wins.

7.4 AWS Organizations for Cost Management

- Consolidated billing = one bill for all accounts + combined volume discounts + shared RIs/Savings Plans across the org.
- Cost allocation tags + linked-account grouping in Cost Explorer show per-team/project spend. SCPs can restrict expensive resource types/regions.
- Separate accounts create natural cost boundaries + per-account budgets. Management account can turn off RI/SP sharing per account if needed.

TIP Multi-account + consolidated billing = volume discounts + isolation + per-team cost visibility with tags.

7.5 SAA-C03 Exam Overview & Domain Review

- 65 questions, 130 minutes, scaled score 100–1000, pass = 720. \$150. Multiple choice + multiple response. Valid 3 years.
- Domains: Design Secure Architectures (30%), Design Resilient Architectures (26%), Design High-Performing Architectures (24%), Design Cost-Optimized Architectures (20%).
- Question style = scenario-based 'best/most cost-effective/most secure' — eliminate wrong answers, watch keywords (cost-effective, highly available, least operational overhead, real-time).

TIP Map keywords to services: 'least operational overhead'→managed/serverless; 'decouple'→SQS/SNS; 'sub-ms'→DynamoDB/DAX; 'archive'→Glacier.

7.6 Practice Exam Sessions

- Take timed full-length practice exams; target 85%+ consistently before the real thing. Review every wrong AND guessed answer.
- Common traps: Multi-AZ (HA) vs Read Replica (scaling); SG (stateful) vs NACL (stateless); Gateway vs Interface endpoint; CloudFront vs Global Accelerator; SQS vs SNS vs Kinesis.
- Practice elimination: two answers are usually clearly wrong; decide between the remaining two on the qualifier (cheapest / most secure / least overhead).

TIP Flag-and-return long questions; read the last sentence first to know what's actually being asked.

7.7 Exam Day Preparation & Career Path

- Logistics: valid ID, stable environment (online proctor) or test center, no notes. Manage time (~2 min/question), answer all (no penalty for guessing).
- Review whitepapers: Well-Architected Framework, AWS pricing, disaster recovery; and FAQs for S3, EC2, VPC, RDS, DynamoDB.

- Career: SAA is foundational for Solutions Architect / Cloud Engineer / DevOps paths. Next steps: SAP (Professional), Security/Networking/Database specialties.

TIP Hands-on labs beat rote memorization — build a VPC, ASG+ALB, and a multi-AZ RDS to cement the concepts.